ELNAZ HEIDARI

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EDUCATION

Ph.D. in Engineering, Clemson University, SC *2022 — 2026*
Dissertation: Explainable physics-guided Transformers for large-scale spatiotemporal flood forecasting.

M.Sc. in Computer Science, Georgia Institute of Technology, GA *2024 — 2026*
Specialization: artificial intelligence and machine learning.

EXPERIENCE

Explainable Transformers for Large-Scale Time-Series Forecasting [Transfer Learning, Physics-guided DL, CUDA, PyTorch, HPC Cluster] **Aug 2022 — Apr 2026**

As a Graduate Research Assistant at Clemson University, developed a Temporal Fusion Transformer framework that forecasts hourly river-flow time series 1–24 hours ahead across 15 U.S. regions, reducing RMSE by 30% compared with NOAA's operational National Water Model. A single model learns from 12 weather time series and 27 catchment attributes, quantifies uncertainty through quantile regression, explains its predictions with attention, and generalizes zero-shot to 10 unseen regions (published in *JAWRA*, 2026; *J. Water Resources Planning and Management*, accepted).

Reinforcement Learning and Probabilistic State Estimation [PPO, Actor-Critic, GAE, MDPs, Q-Learning, SARSA, Kalman Filter, Particle Filter, PyTorch] **Fall 2026**

Implemented PPO from scratch in PyTorch, solving continuous-control LunarLander with a 223.7 mean return (threshold: 200) and running multi-seed ablations. Modeled traffic-signal control as an MDP, where model-based planning had 35% fewer congestion violations than Q-learning. Built Kalman and particle filters for multi-target tracking and satellite localization.

Generative AI: Diffusion Models, GANs, and VAEs [PyTorch, CUDA, Generative Modeling, Image Synthesis] **Spring 2025**

Implemented and benchmarked VAEs, GANs, and diffusion models on MNIST and Fashion-MNIST, comparing sample sharpness, diversity, and training stability: diffusion produced the sharpest, most stable samples, GANs risked mode collapse, and VAEs traded sharpness for diversity.

Distributional Semantics and Word Embeddings [PyTorch, GloVe, CBOW, Skip-Gram, Cosine Similarity] **Fall 2025**

Trained custom Word2Vec models (CBOW and Skip-Gram) on WikiText-2 (2M+ tokens), reaching up to 70% analogy accuracy and 0.74 cosine similarity on semantic relationships.

Language Modeling and Text Generation [PyTorch, NumPy, NLP, RNNs, LSTMs, Transformers, Attention Mechanisms, Gradient Clipping, Positional Encoding] **Fall 2025**

Built RNN, LSTM, and attention-based language models on a custom *Silmarillion* corpus, reducing test perplexity by about 25% through architecture and training improvements, and generated text with temperature-based sampling.

RNNs for Time-Series Forecasting [LSTM, GRU, HPC Cluster] **Aug 2022 — Dec 2025**

Benchmarked LSTM, GRU, and vanilla RNN flood models against NOAA's National Water Model in 4 watersheds, using NOAA reanalysis weather and USGS gauge data (published in *Hydrology*, 2025).

Geospatial Classification from Social Media and Crowd-Sourced Data [Decision Trees, Data Fusion, ArcPy, Python, GitHub] **Summer 2023**

As a Research Assistant at NOAA's National Water Center, built a decision-tree flood classifier that fuses Twitter/X posts and FEMA crowd-sourced reports with NWM inundation maps, improving flood-extent mapping over the NWM alone.

TECHNICAL SKILLS

Languages	Python, C/C++, Go, R, MATLAB, SQL, Bash
ML Frameworks	PyTorch, TensorFlow, Keras, scikit-learn, CUDA, NumPy, Pandas, SciPy, OpenCV
Deep Learning	Transformers, RNNs/LSTMs, CNNs, diffusion models, GANs, VAEs, attention
ML Methods	Time-series forecasting, uncertainty quantification, transfer learning, explainable AI
Reinforcement Learning	PPO, actor-critic, Q-learning, SARSA, value/policy iteration, MDPs
State Estimation	Kalman filters, particle filters, Bayesian filtering
Data & Infrastructure	HPC GPU clusters, AWS (S3, CLI), NetCDF4, Zarr, HDF5, Xarray, GRIB2, ArcPy, Git

SELECTED GRADUATE COURSES

Reinforcement Learning	Fall 2026	**AI for Robotics**	Fall 2026
Machine Learning	Spring 2026	**AI, Ethics, and Society**	Summer 2026
Natural Language Processing	Fall 2025	**Computer Networks**	Summer 2025
Deep Learning	Spring 2025	**Machine Learning for Trading**	Fall 2024
Data Structures and Algorithms	Spring 2024	**ML for Climate Adaptation**	Fall 2023

PUBLICATIONS

- **Heidari, E.**, and Khan, A. A. (2026). Probabilistic Sub-Daily Streamflow Forecasting in Data-Scarce Catchments Through a Zero-Shot Inductive Transfer Learning Approach. *Journal of the American Water Resources Association (JAWRA).*
- **Heidari, E.**, and Khan, A. A. (2026). Interpretable Hourly Flood Forecasting Leveraging Deep Learning Recurrence and Attention Mechanisms. *Journal of Water Resources Planning and Management* (accepted).
- **Heidari, E.**, Samadi, V., and Khan, A. A. (2025). Leveraging Recurrent Neural Networks for Flood Prediction and Assessment. *Hydrology*, 12(4), 90.
- **Heidari, E.**, Samadi, V., and Popescu, I. (2023). Flood Modeling Across Mountainous and Coastal Watersheds: Coupling Data-Driven and Catchment Physical Attributes. *AGU Fall Meeting.*
- **Heidari, E.**, and Samadi, V. (2023). Application of Deep Recurrent Neural Networks for Flood Prediction and Assessment. *EGU General Assembly*, EGU-16678.